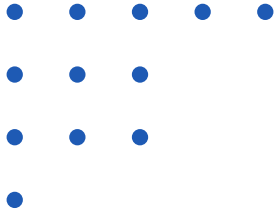


Partitions and Special Numbers

Section 2.8.1–2.8.5

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



$$12 = 5 + 3 + 3 + 1$$

Jeremy Quastel

Today's lecture

1. Partitions

2. Stirling Cycle and Set Numbers

3. Bell Numbers

4. Eulerian Numbers

Section 2.8.1: Integer partitions

Partition of an integer

A partition of a nonnegative integer n is a multiset of positive integers with sum n . Its members are called parts. Let $p(n)$ be the number of partitions of n .

Order does not matter. We normally write the parts in decreasing order. The empty partition gives $p(0) = 1$.

The partitions of 5

$$\begin{aligned}5 &= 5 \\&= 4 + 1 \\&= 3 + 2 \\&= 3 + 1 + 1 \\&= 2 + 2 + 1 \\&= 2 + 1 + 1 + 1 \\&= 1 + 1 + 1 + 1 + 1.\end{aligned}$$

$$p(5) = 7.$$

Partitions and compositions

Composition

A composition of n is an ordered list of positive integers with sum n .

Thus $3 + 2$ and $2 + 3$ are different compositions but the same partition.

For $n \geq 1$, there are 2^{n-1} compositions: choose where to cut a row of n ones, using any subset of the $n - 1$ gaps.

Ferrers diagrams



$$12 = 5 + 3 + 3 + 1$$

Draw one row of dots for each part, left aligned and ordered from longest to shortest.
This represents $12 = 5 + 3 + 3 + 1$.

Conjugate partitions

Conjugation

Interchange rows and columns of a Ferrers diagram. The column lengths form the conjugate partition.

Example

The conjugate of $(5, 3, 3, 1)$ is $(4, 3, 3, 1, 1)$.

Conjugating twice returns to the original partition, so conjugation is a bijection.

A consequence of conjugation

Theorem

The number of partitions of n into exactly k parts equals the number of partitions of n whose largest part is k .

Proof

A Ferrers diagram with k rows becomes a diagram whose longest row has k dots.

Similarly, at most k parts corresponds to largest part at most k .

Partitions with exactly k parts

Notation

Let $p_k(n)$ be the number of partitions of n into exactly k positive parts. Set $p_0(0) = 1$, and use zero for impossible parameters.

Recurrence

For $n, k \geq 1$,

$$p_k(n) = p_{k-1}(n-1) + p_k(n-k).$$

Proof of the partition recurrence

Proof

Separate partitions into two cases.

If a part equals 1, remove one such part. This is a bijection with partitions of $n - 1$ into $k - 1$ parts.

If every part is at least 2, subtract 1 from each of the k parts. This is a bijection with partitions of $n - k$ into k positive parts.

The partition generating function

Identity

$$\sum_{n \geq 0} p(n)x^n = \prod_{j \geq 1} \frac{1}{1 - x^j}.$$

Proof

The factor $1 + x^j + x^{2j} + \dots$ chooses the number of parts equal to j . Multiplying chooses all multiplicities, and the exponent records their sum.

Why the infinite product is well defined

To find the coefficient of x^n , only factors with $j \leq n$ can contribute a positive power.
Therefore each coefficient requires only finitely many factors and finitely many choices.
For largest part at most k , the generating function is the finite product $\prod_{j=1}^k (1 - x^j)^{-1}$.

Exactly k parts: generating function

Identity

$$\sum_{n \geq 0} p_k(n) x^n = \frac{x^k}{(1-x)(1-x^2) \cdots (1-x^k)}.$$

Proof

Subtract 1 from each of k parts, allowing zeros. After discarding zeros, this gives a partition into at most k parts. Conjugation converts this to a partition with largest part at most k .

Distinct parts and odd parts

Euler's partition theorem

The number of partitions of n into distinct parts equals the number of partitions of n into odd parts.

For $n = 5$, distinct-part partitions are 5 , $4 + 1$, $3 + 2$. Odd-part partitions are 5 , $3 + 1 + 1$, $1 + 1 + 1 + 1 + 1$.

A generating-function proof

Proof

Distinct parts contribute

$$\prod_{j \geq 1} (1 + x^j) = \prod_{j \geq 1} \frac{1 - x^{2j}}{1 - x^j}.$$

Cancel the even-indexed denominator factors to obtain

$$\prod_{\substack{j \geq 1 \\ j \text{ odd}}} \frac{1}{1 - x^j},$$

which counts partitions into odd parts. Cancellation is valid coefficient by coefficient.

A direct bijection

Proof

In a partition into odd parts, write the multiplicity of each odd part u in binary. Replace 2^j copies of u by the single part $2^j u$ whenever the corresponding binary digit is 1.

The resulting parts are distinct, because each positive integer has a unique odd part and power of 2. Splitting each distinct part back into copies of its odd part reverses the construction.

A partition example

Example

Start with the odd-part partition

$$3 + 3 + 3 + 1 + 1 + 1 + 1 + 1.$$

Three copies of 3 become $6 + 3$, and five copies of 1 become $4 + 1$. The distinct-part partition is

$$6 + 4 + 3 + 1.$$

Section 2.8.2: Stirling cycle numbers

Definition

The unsigned Stirling cycle number

$$\left[\begin{matrix} n \\ k \end{matrix} \right]$$

counts permutations of $[n]$ with exactly k cycles, including fixed points.

Cycles are unordered, and writing a cycle from a different starting point does not change it.

An example with three elements

Cycles	Count	Permutations
1	2	(123), (132)
2	3	(12)(3), (13)(2), (23)(1)
3	1	(1)(2)(3)

$\begin{bmatrix} 3 \\ 1 \end{bmatrix} = 2,$ $\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3,$ $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 1.$

Boundary values

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1, \quad \begin{bmatrix} n \\ 0 \end{bmatrix} = 0 \ (n > 0), \quad \begin{bmatrix} n \\ n \end{bmatrix} = 1.$$

For $n \geq 1$,

$$\begin{bmatrix} n \\ 1 \end{bmatrix} = (n-1)!.$$

A single cycle is a circular arrangement of all n elements.

The cycle-number recurrence

Recurrence

For $n \geq 1$,

$$\begin{bmatrix} n \\ k \end{bmatrix} = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix} + (n-1) \begin{bmatrix} n-1 \\ k \end{bmatrix}.$$

Distinguish whether n is a one-cycle or belongs to a longer cycle.

Proof of the cycle recurrence

Proof

If (n) is a cycle by itself, the other elements form a permutation with $k - 1$ cycles.

Otherwise remove n from its cycle, leaving a permutation of $n - 1$ elements with k cycles. Conversely, insert n immediately after any of the $n - 1$ elements in its cycle notation.

These disjoint cases give the recurrence.

A small cycle-number table

$n \backslash k$	1	2	3	4	5
1	1				
2	1	1			
3	2	3	1		
4	6	11	6	1	
5	24	50	35	10	1

The row for n sums to $n!$.

Rising factorials

Notation

$$x^{\overline{n}} = x(x+1) \cdots (x+n-1), \quad x^{\overline{0}} = 1.$$

Identity

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k.$$

Proof of the rising-factorial identity

Proof

Write $P_n(x) = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} x^k$. Multiply the recurrence by x^k and sum:

$$P_n(x) = xP_{n-1}(x) + (n-1)P_{n-1}(x).$$

Since $P_0(x) = 1$, induction gives

$$P_n(x) = x(x+1) \cdots (x+n-1).$$

Section 2.8.3: Stirling set numbers

Definition

The Stirling set number

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

counts partitions of the set $[n]$ into k nonempty, unlabeled blocks.

Here the elements being partitioned are distinct. This differs from partitioning the integer n into part sizes.

A set-partition example

The partitions of $\{1, 2, 3\}$ into two blocks are

$$\{\{1\}, \{2, 3\}\}, \quad \{\{2\}, \{1, 3\}\}, \quad \{\{3\}, \{1, 2\}\}.$$

Thus $\left\{ \begin{matrix} 3 \\ 2 \end{matrix} \right\} = 3.$

The order of the blocks and the order of elements within each block do not matter.

The set-number recurrence

Recurrence

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\} + k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}.$$

Proof

If $\{n\}$ is a singleton block, partition the other elements into $k-1$ blocks. Otherwise partition the other elements into k blocks and choose one of these k blocks to receive n .

A small set-number table

$n \backslash k$	1	2	3	4	5
1	1				
2	1	1			
3	1	3	1		
4	1	7	6	1	
5	1	15	25	10	1

Boundary values: $\begin{Bmatrix} 0 \\ 0 \end{Bmatrix} = 1$, $\begin{Bmatrix} n \\ 0 \end{Bmatrix} = 0$ for $n > 0$, and $\begin{Bmatrix} n \\ n \end{Bmatrix} = 1$.

Surjections and set partitions

Identity

The number of onto functions from $[n]$ to $[k]$ is

$$k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}.$$

Proof

The nonempty inverse images of target elements form k blocks. Conversely, partition $[n]$ into k blocks, then assign the k distinct target labels to the blocks.

An explicit formula

Formula

For $n, k \geq 1$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$$

Proof

Use the inclusion–exclusion formula for surjections and divide by the $k!$ ways to label the blocks.

Two-block partitions

Identity

For $n \geq 1$,

$$\left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} = 2^{n-1} - 1.$$

Proof

Choose a nonempty proper subset as one block: $2^n - 2$ choices. Each partition is counted twice, once for either block. Divide by 2.

Powers and falling factorials

Identity

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\underline{k}}.$$

Proof

For a nonnegative integer x , count functions from $[n]$ to $[x]$ by image size k . Partition the domain into its k nonempty fibers, then assign distinct target values to the blocks in $x^{\underline{k}}$ ways.

Since both sides are polynomials agreeing at all nonnegative integers, the identity holds for every x .

Comparing the two Stirling families

	Cycle numbers	Set numbers
Object	Permutations	Set partitions
Parameter k	Number of cycles	Number of blocks
Insertion choices	$n - 1$	k
Row sum	$n!$	B_n

Both recurrences separate off the case where the new element stands alone.

Section 2.8.4: Bell numbers

Bell number

The Bell number B_n counts all partitions of an n -element set. Thus

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\}.$$

$$B_0, B_1, B_2, B_3, B_4, B_5 = 1, 1, 2, 5, 15, 52.$$

A Bell-number recurrence

Recurrence

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k.$$

Proof

In a partition of $[n + 1]$, consider the block containing $n + 1$. Choose the k elements outside that block, then partition those elements in B_k ways. The remaining elements all belong to the distinguished block.

Computing a Bell number

$$\begin{aligned} B_4 &= \binom{3}{0} B_0 + \binom{3}{1} B_1 + \binom{3}{2} B_2 + \binom{3}{3} B_3 \\ &= 1 + 3 + 6 + 5 = 15. \end{aligned}$$

This recurrence computes Bell numbers without first constructing the entire Stirling table.

Exponential generating functions

Definition

The exponential generating function of (a_n) is

$$\hat{A}(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!}.$$

The factorial denominators are useful for objects built on labeled sets. Binomial convolution becomes multiplication of exponential generating functions.

The Bell exponential generating function

Identity

$$\sum_{n \geq 0} B_n \frac{x^n}{n!} = \exp(e^x - 1).$$

Proof

The Bell recurrence implies $\widehat{B}'(x) = e^x \widehat{B}(x)$, with $\widehat{B}(0) = 1$. Solving this differential equation gives $\widehat{B}(x) = \exp(e^x - 1)$.

Dobinski's formula

Formula

For $n \geq 1$,

$$B_n = \frac{1}{e} \sum_{m=0}^{\infty} \frac{m^n}{m!}.$$

Proof

Use $m^n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} m^{\underline{k}}$. For each k ,

$$\sum_{m \geq 0} \frac{m^{\underline{k}}}{m!} = \sum_{m \geq k} \frac{1}{(m-k)!} = e.$$

The finite sum over k then gives eB_n .

Section 2.8.5: Eulerian numbers

Ascent

An ascent of a permutation $a_1 \cdots a_n$ is an index i such that $a_i < a_{i+1}$.

Eulerian number

Write $\left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle$ for the number of permutations of $[n]$ with exactly k ascents.

Counting descents instead gives the same numbers, by reversing the order of a permutation.

An Eulerian example

Ascents	Count	Permutations
0	1	321
1	4	132, 213, 231, 312
2	1	123

$$\left\langle \begin{smallmatrix} 3 \\ 0 \end{smallmatrix} \right\rangle = 1, \quad \left\langle \begin{smallmatrix} 3 \\ 1 \end{smallmatrix} \right\rangle = 4, \quad \left\langle \begin{smallmatrix} 3 \\ 2 \end{smallmatrix} \right\rangle = 1.$$

The Eulerian recurrence

Recurrence

For $n \geq 2$,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = (k+1) \left\langle \begin{matrix} n-1 \\ k \end{matrix} \right\rangle + (n-k) \left\langle \begin{matrix} n-1 \\ k-1 \end{matrix} \right\rangle.$$

Use $\left\langle \begin{matrix} 1 \\ 0 \end{matrix} \right\rangle = 1$, and interpret entries outside $0 \leq k \leq n-1$ as zero.

Insertion positions that preserve ascents

Proof

Start with a permutation of $[n - 1]$ having k ascents and insert the largest element n .

Inserting at the beginning adds no ascent. Inserting in any existing ascent replaces that ascent by one ascent and one descent, preserving the count.

These $k + 1$ positions give the first term of the recurrence.

Insertion positions that add an ascent

Proof

Start instead with a permutation having $k - 1$ ascents. There are n insertion positions in total. Exactly k preserve the ascent count, by the preceding argument.

The remaining $n - k$ positions increase the number of ascents by one: insert at the end or into a descent.

These give the second term. Every permutation of $[n]$ is recovered uniquely by deleting n .

An Eulerian triangle

$n \backslash k$	0	1	2	3	4
1	1				
2	1	1			
3	1	4	1		
4	1	11	11	1	
5	1	26	66	26	1

Each row sums to $n!$, since every permutation has some number of ascents.

Eulerian symmetry

Identity

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \left\langle \begin{matrix} n \\ n-1-k \end{matrix} \right\rangle.$$

Proof

Replace each value a_i by $n+1-a_i$. Every ascent becomes a descent and every descent becomes an ascent. This operation is its own inverse.

The extreme entries are both 1, corresponding to decreasing and increasing permutations.

Eulerian polynomials

Definition

For $n \geq 1$, set

$$A_n(x) = \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle x^k.$$

Recurrence

$$A_n(x) = (1 + (n-1)x)A_{n-1}(x) + x(1-x)A'_{n-1}(x).$$

Multiply the Eulerian-number recurrence by x^k and sum over k .

Power sums as a generating function

Identity

For $n \geq 1$,

$$\sum_{m \geq 0} m^n x^m = \frac{x A_n(x)}{(1-x)^{n+1}}.$$

Proof

For $n = 1$, differentiate the geometric series. Applying $x d/dx$ multiplies the coefficient of x^m by m . Differentiating the right-hand side produces exactly the Eulerian-polynomial recurrence.

An explicit Eulerian formula

Formula

For $0 \leq k \leq n - 1$,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \sum_{j=0}^{k+1} (-1)^j \binom{n+1}{j} (k+1-j)^n.$$

Proof

Multiply the preceding generating-function identity by $(1-x)^{n+1}$ and compare coefficients of x^{k+1} .

An Eulerian calculation

For $n = 4, k = 1$,

$$\left\langle \begin{smallmatrix} 4 \\ 1 \end{smallmatrix} \right\rangle = 2^4 - \binom{5}{1} 1^4 + \binom{5}{2} 0^4 = 16 - 5 = 11.$$

The recurrence gives the same result:

$$\left\langle \begin{smallmatrix} 4 \\ 1 \end{smallmatrix} \right\rangle = 2 \left\langle \begin{smallmatrix} 3 \\ 1 \end{smallmatrix} \right\rangle + 3 \left\langle \begin{smallmatrix} 3 \\ 0 \end{smallmatrix} \right\rangle = 8 + 3 = 11.$$

Practice: identify the family

Question

Which number counts each of the following?

- ① Ways to write 8 as an unordered sum of positive integers.
- ② Permutations of $[8]$ with three cycles.
- ③ Partitions of $[8]$ into three nonempty blocks.
- ④ All partitions of $[8]$.
- ⑤ Permutations of $[8]$ with three ascents.

Practice: answers

Integer partitions	$p(8)$
Permutations with three cycles	$\left[\begin{array}{c} 8 \\ 3 \end{array} \right]$
Set partitions into three blocks	$\left\{ \begin{array}{c} 8 \\ 3 \end{array} \right\}$
All set partitions	B_8
Permutations with three ascents	$\left\langle \begin{array}{c} 8 \\ 3 \end{array} \right\rangle$

- Integer partitions are counted by products recording part multiplicities.
- Stirling cycle and set numbers distinguish cycles from blocks.
- Bell numbers sum over all possible numbers of blocks.
- Eulerian numbers count ascent patterns and appear in power-sum generating functions.

Course Review